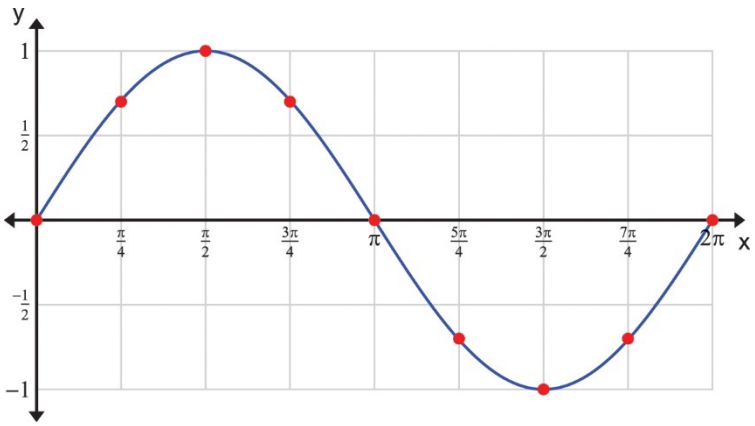
Graph $y = \sin x$ and $y = \cos x$

The table gives several values of x and the corresponding values of $\sin x$.

x	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\sin x$	0	$\frac{\sqrt{2}}{2}$	1	$\frac{\sqrt{2}}{2}$	0	$-\frac{\sqrt{2}}{2}$	-1	$-\frac{\sqrt{2}}{2}$	0

Plotting these as ordered pairs gives the graph of $y = \sin x$.



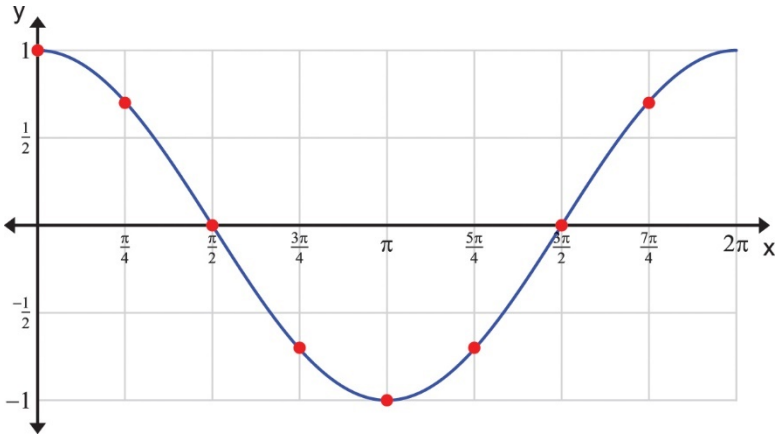
The sine function is periodic with period 2π.

This means that the pattern shown between 0 and 2π repeats infinitely far to the left and right.

The next table gives several values of x and the corresponding values of $\cos x$.

x	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\cos x$	1	$\frac{\sqrt{2}}{2}$	0	$-\frac{\sqrt{2}}{2}$	-1	$-\frac{\sqrt{2}}{2}$	0	$\frac{\sqrt{2}}{2}$	1

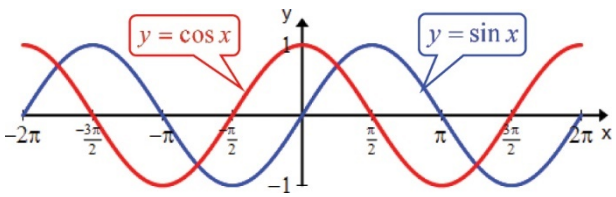
Plotting these as ordered pairs gives the graph of $y = \cos x$.



The cosine function is also periodic with period 2π.

The pattern shown between 0 and 2π repeats infinitely far to the left and right.

The graphs of $y = \sin x$ and $y = \cos x$ are shown on the interval $[-2\pi, 2\pi]$ for comparison.



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Characteristics of the graphs of sin *x* and cos *x*: The domain is $(-\infty, \infty)$. The range is $[-1, 1]$. The period is 2π.

The graph of sin *x* is symmetric with respect to the origin.
$y = \sin x$ is an odd function.

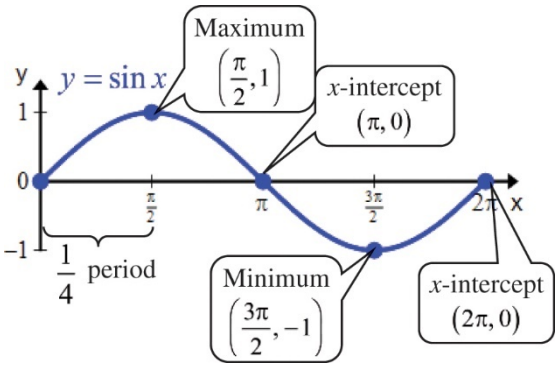
The graph of cos *x* is symmetric with respect to the *y*-axis.
$y = \cos x$ is an even function.

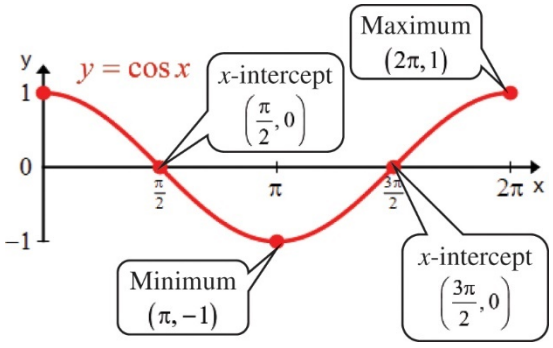
The graphs of sin *x* and cos *x* differ by a horizontal shift of $\frac{\pi}{2}$.

Graph $y = A\sin x$ and $y = A\cos x$

To graph variations of these functions we may graph the key points - relative minima, relative maxima, and *x*-intercepts.

The graphs of both the sine and cosine function alternate between a relative minimum or maximum and an *x*-intercept for each quarter of a period.





The graph of $y = A \cdot f(x)$ is the graph of $y = f(x)$ with

a vertical stretch if $|A| > 1$

a vertical shrink if $0 < |A| < 1$

a reflection across the *x*-axis if $A < 0$

Amplitude: For the graphs of $y = |A|\sin x$ and $y = |A|\cos x$, the vertical scaling factor $|A|$ is the amplitude.
The amplitude represents the amount of deviation from the central position of the sine wave.
The amplitude is half the distance between the maximum value of the function and the minimum value of the function.

Identify the amplitude of each function.

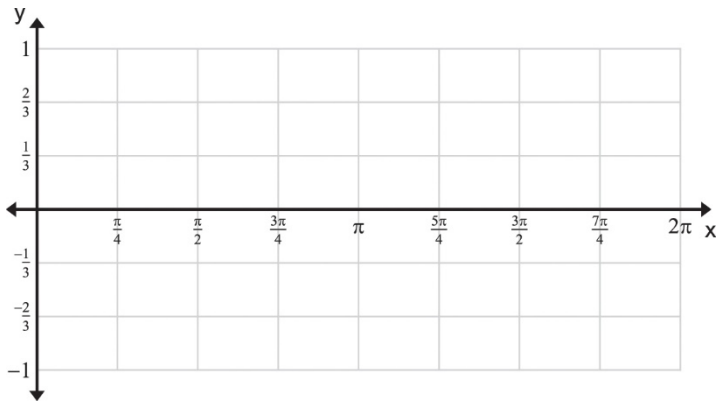
1. $f(x) = -8\sin x$

2. $g(x) = \frac{2}{3}\cos x$

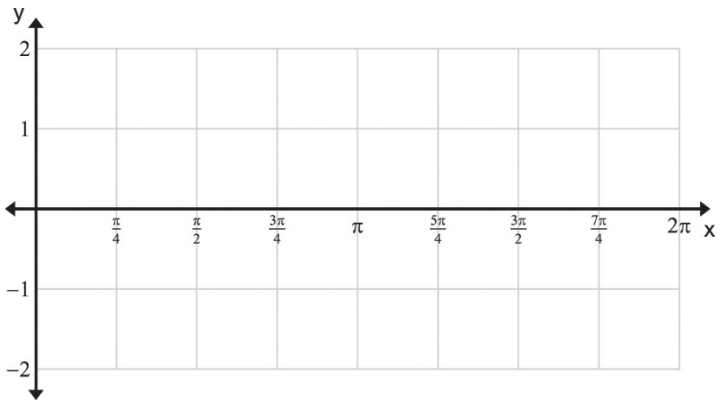
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Graph the function.

3. $y = \frac{2}{3}\cos x$

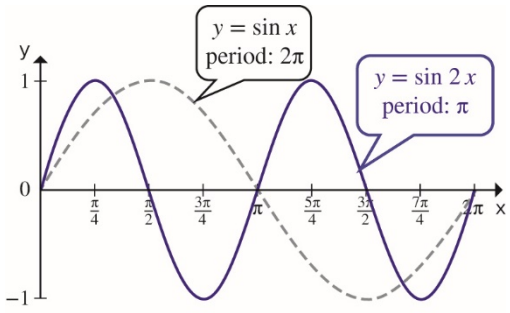


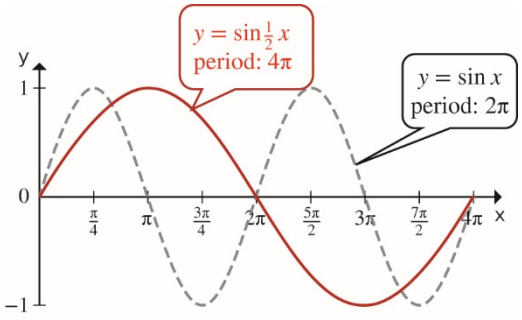
4. $y = -2\sin x$



Graph $y = A\sin Bx$ and $y = A\cos Bx$

The graph of $y = f(Bx)$ is the graph of $y = f(x)$ with a horizontal shrink or stretch.
For $y = \sin x$ and $y = \cos x$, the factor B will affect the period of the graph.





To find the interval for one complete cycle of the graphs of $y = \sin Bx$ and $y = \cos Bx$ for $B > 0$, solve the inequality.

$$0 \le Bx \le 2\pi$$

$$\frac{0}{B} \le \frac{Bx}{B} \le \frac{2\pi}{B}$$

$$0 \le x \le \frac{2\pi}{B}$$

Start value for the cycle — End value for the cycle

For $B > 0$, the period of $y = \sin Bx$ and $y = \cos Bx$ is $\frac{2\pi}{B}$.

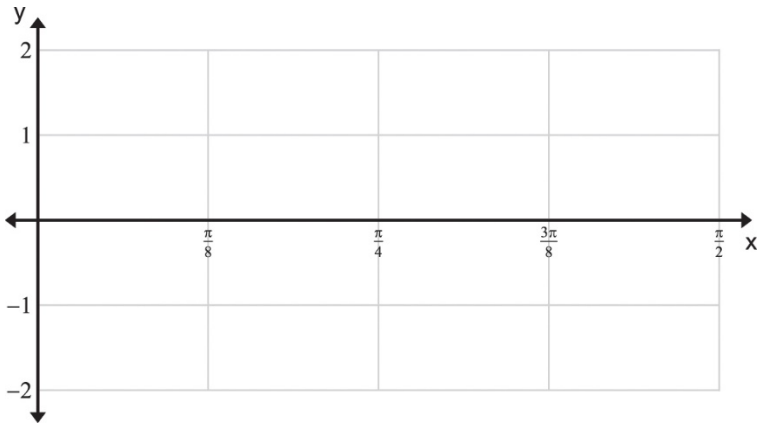
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To analyze the period of a sine or cosine function with a negative coefficient on x, for $B > 0$:

Rewrite $y = \sin(-Bx)$ as $y = -\sin Bx$ Rewrite $y = \cos(-Bx)$ as $y = \cos Bx$

Graph the function over one full period and identify the amplitude and period.

5. $y = 2\cos(-4x)$



Graph $y = A\sin(Bx - C) + D$ and $y = A\cos(Bx - C) + D$

The graph of $y = f(x - h)$ is the graph of $y = f(x)$ with a horizontal shift of $|h|$ units. If $h > 0$ the shift is to the right, and if $h < 0$ the shift is to the left.

The graphs of $y = A\sin(Bx - C)$ and $y = A\cos(Bx - C)$ have both a change in period and a horizontal shift. The horizontal shift to the left or right is called the phase shift.

For $B > 0$, the phase shift can be determined by solving the following inequality.

$$0 \le Bx - C \le 2\pi$$

$$C \le Bx \le 2\pi + C$$

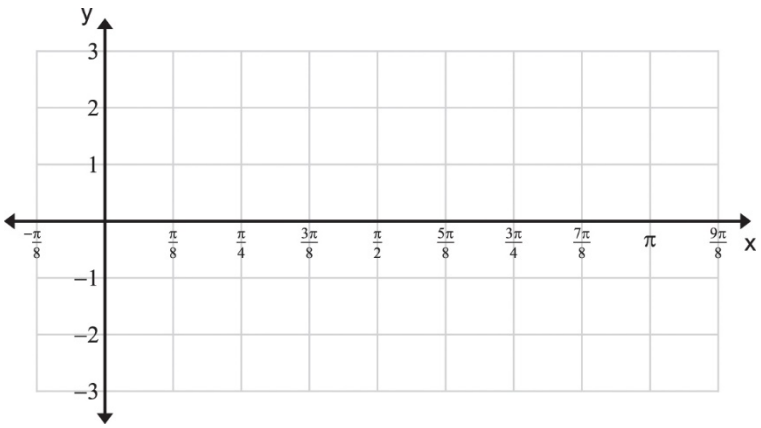
The phase shift is the left endpoint.

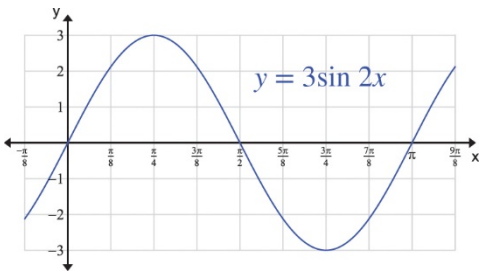
$$\frac{C}{B} \le x \le \frac{2\pi + C}{B}$$

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6. Graph the function over one full period.
Identify the amplitude, period, and phase shift.

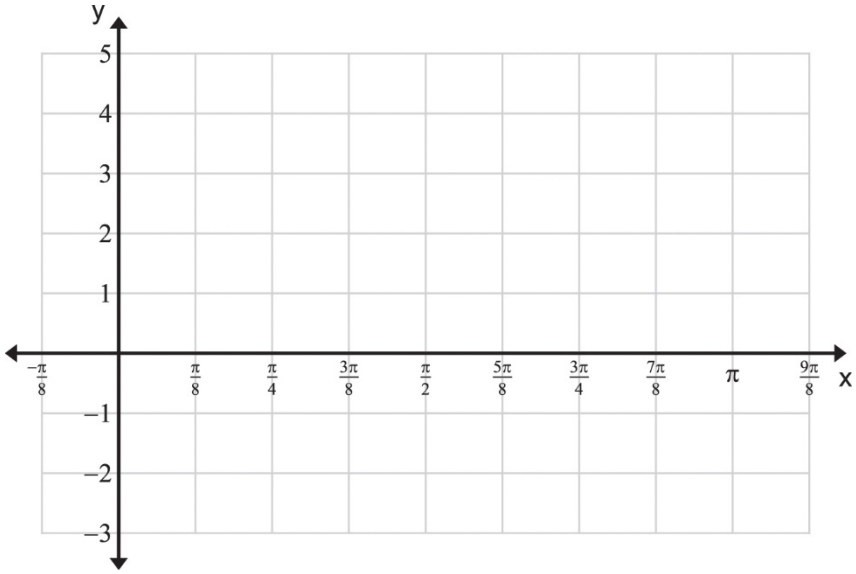
$$y = 3\sin\left(2x - \frac{\pi}{4}\right)$$





To find the vertical shift, recall that the graph of $y = f(x) + D$ is the graph of $y = f(x)$ shifted $|D|$ units upwards for $D > 0$ and $|D|$ units downwards if $D < 0$.

7. Graph $y = 3\sin\left(2x - \frac{\pi}{4}\right) + 2$ (range?)



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Properties of the General Sine and Cosine Functions

For the graphs of $y = A\sin(Bx - C) + D$ and $y = A\cos(Bx - C) + D$ with $B > 0$.

Amplitude = $|A|$ Period = $\frac{2\pi}{B}$ Phase shift = $\frac{C}{B}$ Vertical shift = D

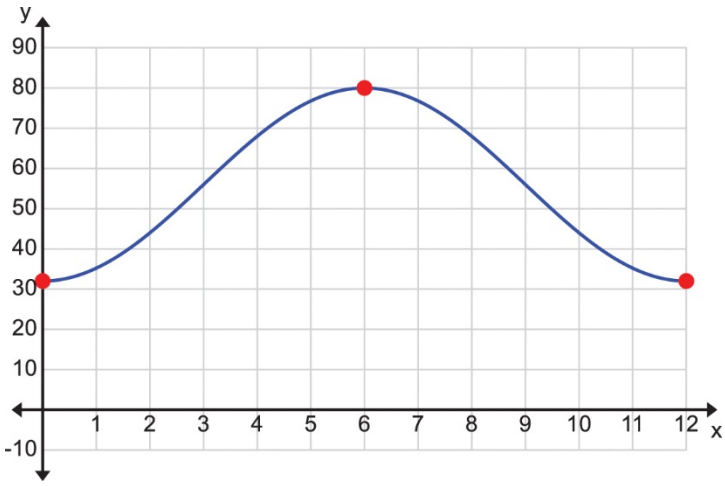
One full cycle is given on the interval $0 \le Bx - C \le 2\pi$.

Domain is $-\infty < x < \infty$.
Range is $-|A| + D \le y \le |A| + D$.

Model Sinusoidal Behavior

8. The monthly high temperature for Buffalo, New York, peaks at an average high of 80° in July down to an average high of 32° in January. Assume that this pattern for monthly high temperatures continues indefinitely and behaves like a cosine wave.

   Write a function of the form $H(t) = A\cos(Bt - C) + D$ to model the average high temperature.
   The value $H(t)$ is the average high temperature for month t, with January as $t = 0$.



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